

LAB 2 [Group # → 9 to 16]

[Platform → HACKERRANK]

Introduction To Programming

(17-01-22)

SOLUTIONS

Q1 → FIND DIVISORS

```
n = int(input())
## input has been taken
## Now loop would be run to find all
## the successive divisors

for num in range(1,n+1):
    ##Why n+1? -> because 'range(1,n)' runs from 1 to (n-1)
    if (n%num) == 0:
        ## that is num divides n
        print(num)
```

Q2 → CAN SAM PLAY?

```
d = int(input())
## Took the day as integer input
t = int(input())
## Took time in minutes as integer input

answer = 'NO'
## initialized answer with 'NO'
## Not a necessary step, but to ensure the
## flow of logic is clear

if (d==6 or d==0):
    ## if day is either saturday or sunday
    if t<=240:
        answer = 'YES'
    else :
        ## that is t > 240
        answer = 'NO'
    ## outer if ends
else:
    ## d is among 1 to 5
    if t<=120:
        answer = 'YES'
    else:
        answer = 'NO'
    ## outer else ends

print(answer)
```

Q3 FIND MY BILL

```
bill = 0
## initialized variable bill with 0
num_of_units = int(input())
## how many units I've used so far

if (num_of_units <= 150):
    bill = 0

elif (num_of_units > 150) and (num_of_units <= 250):
    bill = (num_of_units - 150) * 10

else:
    bill = 1000 + (num_of_units - 250) * 15

print(bill)
```

Q4 WHAT'S YOUR BONUS

```
service_year = int(input())
## number of years, one has worked
## in the company
salary = int(input())
## salary of the employee

## based on the logistics mentioned
if service_year > 8:
    bonus=15/100*salary

elif service_year >=4 and service_year <=8:
    bonus = 10/100*salary

else:
    ## less than 4 years of experience
    bonus = 5/100*salary

print(int(bonus))
```

Q5 FIND X

```
count_even = 0
count_odd = 0

n = int(input())
##Taken input for the number of values
## I have used while loop, for loop will work the same
i= 0
## i will be used to test when to stop the while loop

while(i<n):
    num = int(input())
    ## take input for the number
    if num%2 == 0:
        ## num is even
        count_even = count_even + 1
    else:
        ## num is odd
        count_odd = count_odd +1

    i = i+1 ## loop condition updated
## while loop has ended

print(count_even - count_odd)
```

Q6 FIBONACCI SEQUENCE [NOT-GRADED]

```
a1 = 0
a2 = 1
## First two values initialized as mentioned

n = int(input())
## input has been taken
▼ if n==1:
    ## base condition 1
    temp = 0
▼ if n==2:
    ## base condition 2
    temp=1
## if none of the two base cases exist
## let's proceed for n>2

▼ for x in range(3, n+1):
    ## loop runs starting from 3
    fibo_value = a1+a2
    a1=a2
    a2=fibo_value

print(fibo_value)
```

Q7 LARGEST TWO [NOT GRADED]

```
max1 = 0
## the 1st largest number
max2 = 0
## the 2nd largest number
n = int(input())

for i in range(n) :
    ## loop runs for n times
    num = int(input())
    if num > max1 :
        ## implies num is largest of all the inputs
        ## till found
        max2 = max1
        ## 2nd largest has been updated with first largest
        max1 = num
        ## first largest has been updated with num

    elif num > max2 :
        ## num is the new second largest
        max2 = num
        ## max2 has been updated with num

print(max2, max1)
```
